

AgriSense: Industry and Project Case Study

1. Overview and Framing

This case study looks at the industry context for my live brief project, AgriSense, which I developed in response to a brief from TeleAgriCulture. TeleAgriCulture is an open platform that lets people share environmental sensor data from IoT boards, and the brief asked me to build a refined AI and data workflow on top of an existing notebook pipeline they had developed at the Schmiede festival in Salzburg.

The sector I am looking at is smart agriculture and community IoT sensing. I chose this focus because it is directly relevant to the project and because it sits in an interesting space between commercial technology development and community or artistic practice. My project ended up being a full-stack tool that pulls sensor data from the TeleAgriCulture API, classifies it using a decision tree model, and shows the results in a dashboard with a forecast and alert system on top.

The brief came through V2_Lab for the Unstable Media in Rotterdam, which is a long running arts and technology organisation that works with practitioners on creative computing projects. The project and this case study are connected in that the decisions I made during development, for example choosing an interpretable model over something more complex were shaped by the industry context and what the partner actually needed.

2. Industry and Sector Context

The smart agriculture technology sector is quite large and has grown a lot in recent years. There are commercial companies like John Deere and Trimble who make precision agriculture tools for large farms, and these tend to use complex AI models and proprietary data systems and then there is the smaller but growing community of open source platforms that are more aimed at smaller growers, urban agriculture projects and researchers.

Sitting at the latter end of that category is TeleAgriCulture; describing itself as a community platform that facilitates access to environmental sensing technologies for anyone interested in exploring the interactions between plants, animals, humans and their environments (TeleAgriCulture, 2023) means the intended users are not professional engineers and the tools need to be accessible and understandable rather than just technically powerful.

A research paper I found looked at smart farming technologies and found explored the recent big increase in IoT and AI research in agriculture between 2020 and 2024, but noted that many of the solutions developed are too expensive or technically complex for smallholder farmers (Miller et al., 2025). This feels relevant to my project because it suggests there is a real gap between what gets built and what community users actually need.

In terms of roles and workflows in this sector, people tend to need a mix of hardware knowledge (working with microcontrollers and sensors), software skills (APIs, data processing, front end development) and some understanding of the context they are working in, which in this case is agriculture and environmental monitoring. The work tends to be commissioned through arts organisations, research institutions and/or community initiatives rather than through traditional commercial contracts.

TeleAgriCulture for example works through residencies, workshops and community collaboration with the output of projects often being shared publicly and documented as part of the work.

There are organisations doing similar things, for example FarmOS is an open source farm management platform and there are various academic and civic IoT projects around air quality and urban environment sensing. What is different about TeleAgriCulture is the emphasis on combining data sensing with creative and artistic practice, which is where V2_ and the brief come in.

3. Situated Case Analysis

The main focus of my analysis here is the partner organisation and the brief context, and specifically the decision to use interpretable AI models rather than more complex ones.

The brief was quite specific about this. It used the phrase ‘microdosing AI’ meaning small-scale and interpretable AI that is understandable by non-experts. This is quite different from how AI tends to be used in commercial agriculture, where models like deep neural networks are common because they optimize for prediction accuracy (Fuentes-Penailillo et al., 2024). Choosing a shallow decision tree with a maximum depth of four was a response to the brief rather than following the default of the field. The advantage is that you can read the exact rules the model is using, which matters when the users are community growers who might not trust or understand a black box output.

In practice there were several real constraints that shaped how the project developed. The TeleAgriCulture API uses cursor-based pagination and the data returned is not always consistent between different boards and sensors, which meant a lot of time went into making the data fetch and cleaning stages robust. The capacitive soil moisture sensor I was using also outputs raw ADC values rather than a calibrated percentage, which I did not initially realise, so I had to add a calibration step to the pipeline. These kinds of messy data issues are common in real IoT projects but they are not usually talked about much in tutorials or academic papers.

Another constraint was thinking about who would actually use the tool and how. Because the brief has a community focus and involves collaborators at institutions like Minodu in Berlin and Togo, I decided early on that the tool should be accessible through a web browser without needing to install anything. This influenced the decision to build a FastAPI backend and React frontend rather than keeping everything in a Jupyter notebook. A notebook is fine for demonstrating the pipeline but it requires Python knowledge to run, which most community growers would not have.

There were also time constraints. I was building this alongside other coursework, which meant some parts of the project are more developed than others. The alert system and forecast feature work, but the peer benchmarking could be more sophisticated, for example by filtering peers by geographic region or sensor type rather than just sensor name. This is something I would develop further with more time.

Looking back at the decisions evidenced in my portfolio, the choice of tech stack, the pipeline architecture, and the UI design all reflect the brief's emphasis on accessibility and interpretability. The dashboard shows the decision tree rules directly to the user, which is an unusual choice but one that I think is right for this context.

4. AI-Assisted Practice and Future Implications

AI tools are currently used in the agriculture sector mainly for things like yield prediction, disease detection from images, and irrigation control. A review from 2025 found that AI-enabled models can improve yield prediction by around 20% and that smart irrigation systems can improve crop productivity by 25% (Padhiary et al., 2025). These are significant gains but they tend to rely on large datasets, expensive infrastructure, and technical expertise that most small or community-scale growers do not have access to.

In the kind of creative computing context this project sits in, there are different tensions around AI. One is the question of interpretability and trust. When a tool is making recommendations to a community grower about whether to irrigate or whether there is frost risk, the grower needs to be able to understand why the system is saying that, not just accept an output. This is one reason the decision tree approach is well suited here even though it is less accurate than a neural network would be.

There are also questions around who AI systems are built for and whether they replicate existing power imbalances. If a sensor classification system makes decisions that farmers cannot interrogate or challenge, it reproduces the dependency on technical experts and commercial products that smallholder farmers already experience. The brief's emphasis on interpretable AI can be read as a response to this, and it is something I tried to carry through in the design of the tool.

One conclusion I would draw is that interpretable machine learning approaches will become more important in community facing civic technology as awareness of accountability and explainability issues grows. A second conclusion is that the role of the creative computing practitioner in this kind of context involves translating between technical systems and community understanding, which is a distinct skill that is not always recognised in job descriptions or training programmes.

A plausible near future scenario is that platforms like TeleAgriCulture develop federated approaches where boards contribute to a shared model without sending all their data to a central server. This would improve classification accuracy across the network while preserving some privacy, and would make the peer benchmarking aspect of the tool more meaningful as the number of boards grows.

5. Personal Positioning and Next Steps

Working on this project has given me a clearer sense of what kind of work I find interesting, which is building technical tools that are designed for non-technical users, particularly in community or civic contexts. I am less drawn to commercial enterprise software development and more interested in the kind of hybrid technical and cultural role that organisations like V2_, Nesta, or the Open Data Institute represent.

I am now aiming toward placement and graduate roles in software engineering, data science, and applied technical development, particularly in environments where systems, data analysis (or humanities) and users meet. This includes areas such as digital public services, simulation systems, and data-driven product teams. The common thread is building reliable tools that sit behind real services, whether that is supporting decision making, improving accessibility, or maintaining system stability.

Two clear next steps come out of this. First, I want to strengthen my backend and systems skills, particularly around APIs, databases, and scalable Python-based services, alongside more structured experience with SQL in production-style environments. Second, I am actively targeting placement applications where I can work in teams building and maintaining real systems, with a focus on roles that involve data pipelines, internal tools, or user-facing platforms. I am also continuing to develop my portfolio so that projects like AgriSense are presented more clearly as production-style systems rather than standalone coursework, with better documentation and deployment examples.

References

Fuentes-Penailillo, F., Gutter, K., Vega, R. and Silva, G.C. (2024) Transformative technologies in digital agriculture: leveraging Internet of Things, remote sensing, and artificial intelligence for smart crop management. Journal of Sensor and Actuator Networks, 13(4), p.39.

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Padhiary, M. et al. (2025) Revolutionizing agriculture: a review of smart farming technologies for a sustainable future. Discover Applied Sciences. [Online] Available at: <https://link.springer.com/article/10.1007/s42452-025-07561-6> [Accessed 19 March 2026].

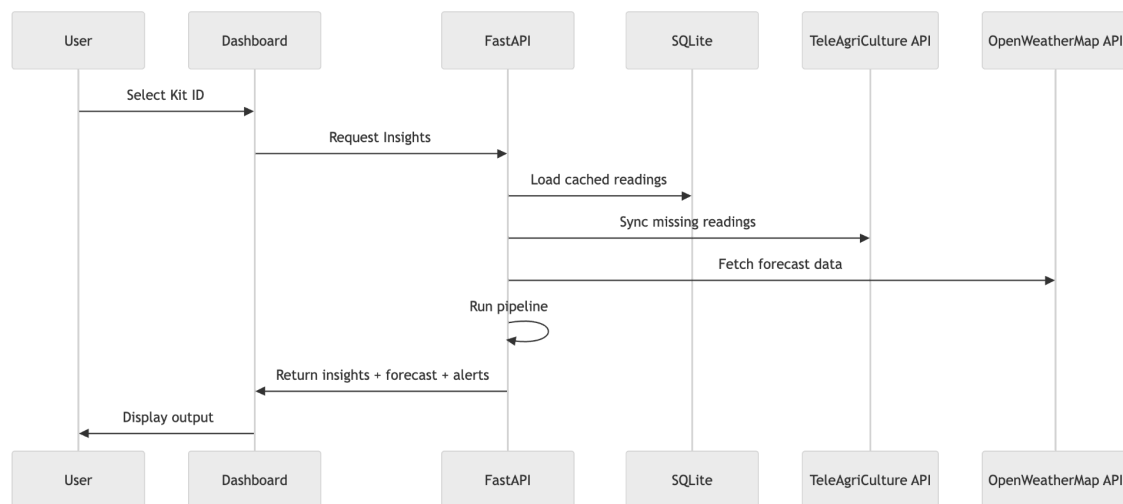
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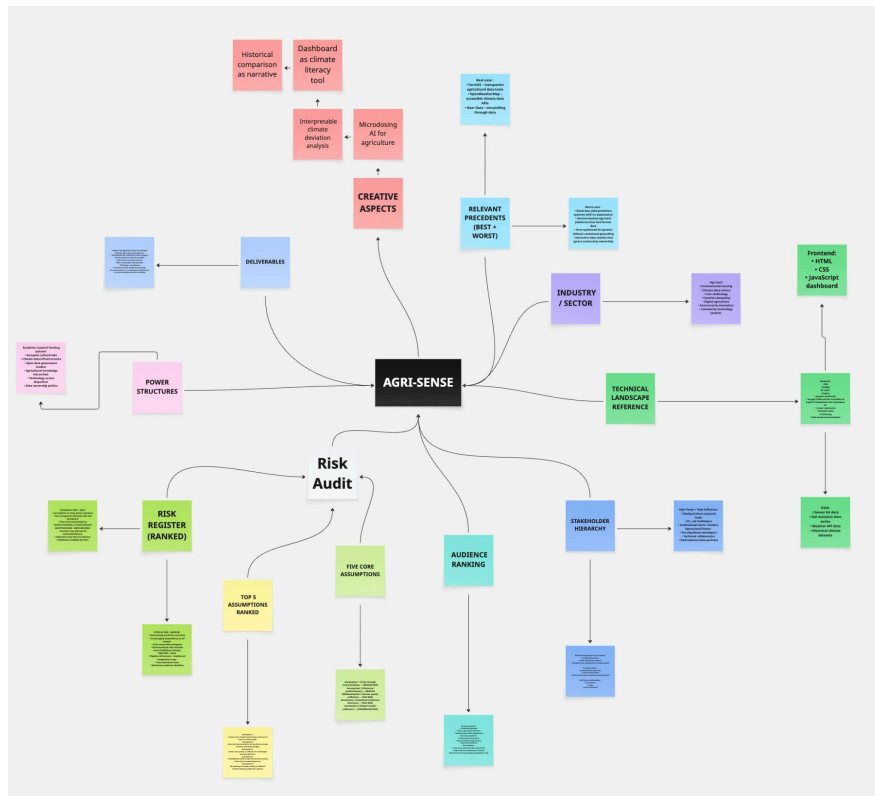
Visual Appendix

Canva visual presentation: <https://canva.link/mil18rr6u1ck3d8>

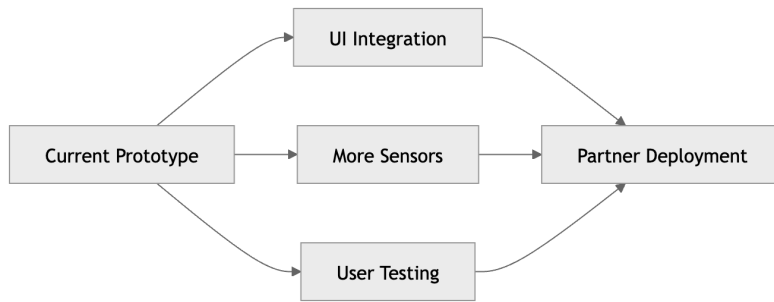
A. System Architecture Overview



B. Initial Miro board and final reflection future flowchart



https://miro.com/app/board/uXjVG0h3dQg=?share_link_id=677175892638



C. Dashboard Screenshot

